Lecture plan · for the lecturer

Attention and transformers

AI · University · Lecture 3

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Compute scaled dot-product attention.
- Explain √d and position.
- Sequence the transformer for novices.

Activity: One head by hand

1. Warm-up (5 min): أيّ اسم؟
2. Compute (20 min): Pairs compute QK^T, scale, softmax and the weighted values for three 2-D tokens.
3. Break (10 min): Repeat with d = 512 random vectors without scaling and look at the weights.
4. Build (20 min): Add heads, a residual path and positions, each with the problem it solves.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- What happened to the weights without √d?
- Can attention maps explain a prediction?
- Which part will you teach last?

Exit ticket

- Write $\text{softmax}(QK^T/\sqrt{d})V$ and label what each letter means.

Try it in your next lecture
Give pairs three 2-D query and key vectors; they compute scores, softmax and output by hand, then compare with three lines of NumPy.

Seminar sheet · for students

Attention and transformers

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Name _____ Class _____ Date _____

1 **Attention alone treats inputs as a set. What makes word order matter?**

◯ More attention heads ◯ Positional encodings ◯ A larger vocabulary

2 **Compute one attention row. A query scores the three keys 2, 0 and 0 after scaling.**

With $e^2 \approx 7.4$, the softmax weight on the first key is about _____ %, and on each of the others about _____ %.

Word bank: 79 · 50 · 33 · 11

3 **Which statements about self-attention are true? Choose all that apply.**

☐ Its cost grows with the square of sequence length

☐ Without position information it ignores word order ☐ Each head's weights in a row sum to 1

☐ It needs recurrence to see distant tokens

4 **Dividing by $\sqrt{d}$ is mainly there to make the computation faster.**

◯ True ◯ False

5 **Match each component to its job.**

Positional encoding	• • • •	Stops a token seeing future tokens
Multiple heads	• • • •	Keeps gradients flowing in deep stacks
Residual connection	• • • •	Capture different relations
Causal mask	• • • •	Injects word order

✎ **What is the minimum a student must compute by hand before using a transformer library in your course?**